

Setting up R and RStudio

Learning objective

- To be able to access R and RStudio, either through RStudio.cloud or by downloading and installing these softwares to your computer.

Introduction

- To begin your R journey, we'll first set up the necessary software: **R** and **RStudio**.
- R is the programming language you'll use to write your code, while **RStudio** is an integrated development environment (IDE) that provides a user-friendly interface for working with R more efficiently.
- At its simplest, R is like a car's **engine** while RStudio is like a car's **dashboard** as illustrated

R: Engine



RStudio: Dashboard



R Platforms

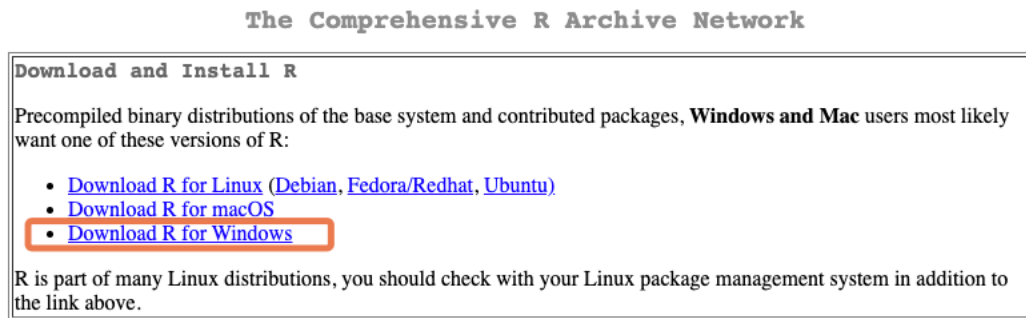
- There are two main ways that you can access and work with **R** and **RStudio**:
 - *download them to your computer, or*
 - *use a web server to access them on the cloud.*
- Using **R** and **RStudio** on the cloud is the less common option, but it may be the right choice if you are just getting started with programming, and you do not yet want to worry about installing software. You may also prefer the cloud option if your local computer is old, slow, or otherwise unfit for running R.
- Below, we outline the setup process for **RStudio Cloud**, **RStudio on Windows**, and **RStudio on macOS** separately.
- For this training, we will be using **RStudio for Windows**

Set up on Windows OS

Download and install R

If you're working on Windows, follow the steps below to download and install R:

1. Go to cran.rstudio.com to access the R installation page. Then click the download link for Windows:



2. Choose the “base” sub-directory.

R for Windows	
Subdirectories:	
base	Binaries for base distribution. This is what you want to install R for the first time .
contrib	Binaries of contributed CRAN packages (for R $\geq$ 4.0.x).
old contrib	Binaries of contributed CRAN packages for outdated versions of R (for R $<$ 4.0.x).
Rtools	Tools to build R and R packages. This is what you want to build your own packages on Windows, or to build R itself.

3. Then click on the download link at the top of the page to download the latest version of R:

R-4.5.1 for Windows

[Download R-4.5.1 for Windows](#) (86 megabytes, 64 bit)

[README on the Windows binary distribution](#)

[New features in this version](#)

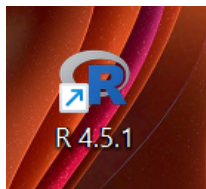
Note that the screenshot above may not show the latest version.

4. Installation steps

- *Once the download is complete, locate and click on the downloaded file.*
- *Follow the instructions that appear in the installation pop-up window.*
- *During the installation process, there's no need to change any default settings.*
- *Simply click “**Next**” until the installation is complete.*

Congratulations, you have successfully installed R in your computer.

You should now be able to see this icon in your computer



Great! You should now have R on your computer. However, you are likely not going to interact with R directly.

Instead you'll use the **RStudio IDE to work with R**.

Download, install RStudio

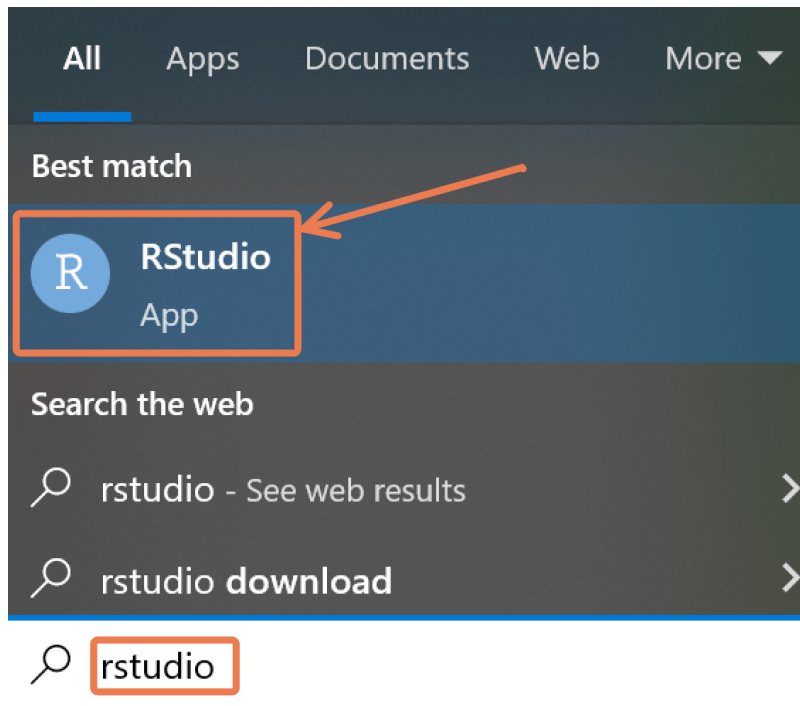
- To download RStudio, go to rstudio.com/products/rstudio/download/#download and download the Windows version by clicking on the link shown.

2: Install RStudio

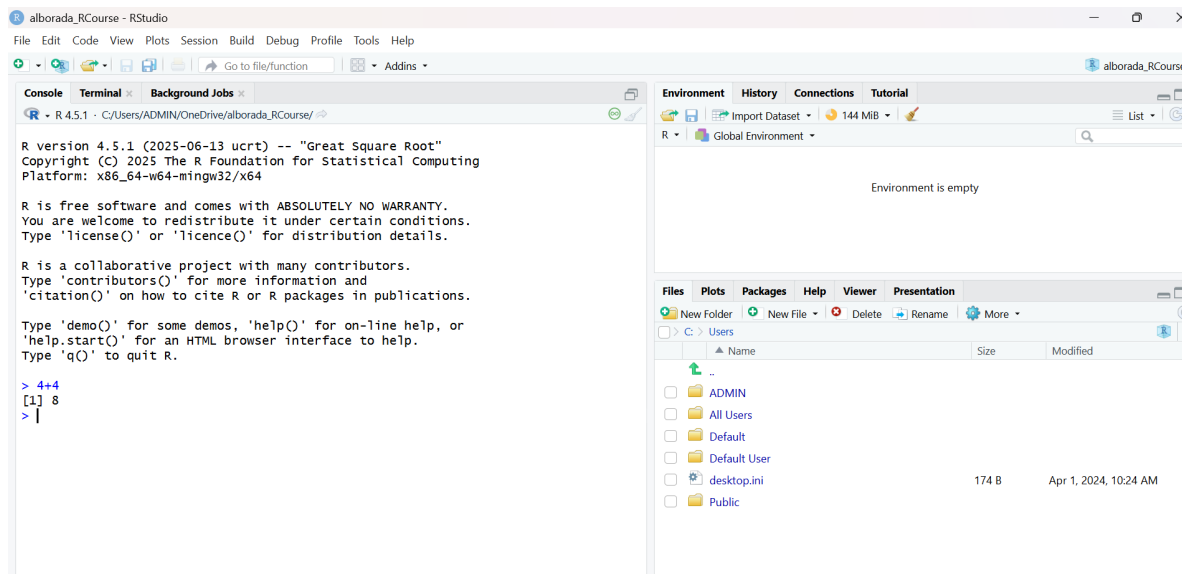
DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS

Size: 287.98 MB | [SHA-256: BCA050AD](#) | Version:
2025.09.1+401 | Released: 2025-10-01

- After the download is finished, click on the downloaded file and follow the installation instructions.
- Once installed, RStudio can be opened like any application on your computer:
 - *you can click on the Windows key to bring up the Start menu, and search for “rstudio” (depending on your windows OS version).*
 - *click to to open the app:*



- You should see a window like this:



- This is RStudio, your new home for a long time to come!
- You can start using R by typing code into the “console” pane on the left:

```
R version 4.1.2 (2021-11-01) -- "Bird Hippie"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> 2+2
[1] 4
> 4+4
```

Write R code here!

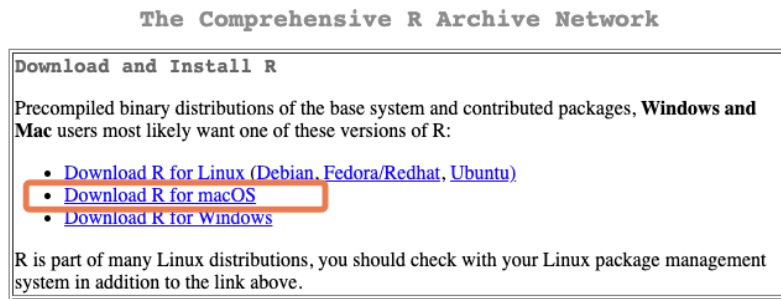
- Try using R as a calculator here; type `2 + 2` and press Enter.
- That's it; you're ready to roll. Proceed to the “wrapping up” section of the lesson.

Set up on macOS

Download and install R

If you're working on macOS, follow the steps below to download and install R:

1. Go to cran.rstudio.com to access the R installation page. Then click the link for macOS:



2. Download and install the relevant R version for your Mac. For most people, the first option under “Latest release” will be the one to get.

[R 4.2.0.pkg](#) (notarized and signed)
SHA1 hash: 2a90b8629a447229a8946a9a9e71564587d7
(ca. 90MB) for Intel Macs

Latest version for Intel Macs

[R 4.2.0-arm64.pkg](#) (notarized and signed)
SHA1 hash: a6c29c2a25194d3169f12d4548263ec28a8ff
(ca. 89MB) for M1 Macs only!

Latest version for M1 Macs

[NEWS](#) (for Mac GUI)

[Mac-GUI-1.78.tar.gz](#)
SHA1 hash: 230aef1b70b71660f6504675c5782792a6db2bc

Note: Previous R versions for El Capitan can be found in the [el-capitan/base](#) directory.

[R 3.6.3.nn.pkg](#) (signed)
SHA1 hash: c4c2c9b1f9a45d778f03b8a9a25a9123b3551e4
(ca. 77MB)

For older macs

Latest release:

R 4.2.0 binary for macOS 10.13 (**High Sierra**) and higher, **Intel 64-bit** build, signed and notarized package. Contains R 4.2.0 framework, Rapp GUI 1.78 in 64-bit for Intel Macs, Tcl/Tk 8.6.6 X11 libraries and Texinfo 6.7. The latter two components are optional and can be omitted when choosing "custom install", they are only needed if you want to use the `teletk` R package or build package documentation from sources.

Note: the use of X11 (including `teletk`) requires [XQuartz](#) to be installed (version 2.7.11 or later) since it is no longer part of macOS. Always re-install XQuartz when upgrading your macOS to a new major version.

This release supports Intel Macs, but it is also known to work using Rosetta2 on M1-based Macs. For native Apple silicon arm64 binary see below.

Important: this release uses Xcode 12.4 and GNU Fortran 8.2. If you wish to compile R packages from sources, you may need to download GNU Fortran 8.2 - see the [tools](#) directory.

R 4.2.0 binary for macOS 11 (**Big Sur**) and higher, **Apple silicon arm64** build, signed and notarized package. Contains R 4.2.0 framework, Rapp GUI 1.78 for Apple silicon Macs (M1 and higher), Tcl/Tk 8.6.12 X11 libraries and Texinfo 6.8.

Important: this version does NOT work on older Intel-based Macs.

Note: the use of X11 (including `teletk`) requires [XQuartz](#) (version 2.8.1 or later). Always re-install XQuartz when upgrading your macOS to a new major version.

This release uses Xcode 13.1 and experimental GNU Fortran 12 arm64 fork. If you wish to compile R packages which contain Fortran code, you may need to download GNU Fortran for arm64 from <https://mac-r-project.org/tools>. Any external libraries and tools are expected to live in `/opt/R/arm64` to not conflict with Intel-based software and this build will not use `/usr/local` to avoid such conflicts (see the [tools page](#) for more details).

News features and changes in the Rapp Mac GUI

Sources for the Rapp GUI 1.78 for macOS. This file is only needed if you want to join the development of the GUI (see also [Mac-GUI repository](#)), it is not intended for regular users. Read the INSTALL file for further instructions.

Binaries for legacy OS X systems:

R 3.6.3 binary for OS X 10.11 (El Capitan) and higher, signed package. Contains R 3.6.3 framework, Rapp GUI 1.70 in 64-bit for Intel Macs, Tcl/Tk 8.6.6 X11 libraries and Texinfo 5.2. The latter two components are optional and can be omitted when choosing "custom install", they are only needed if you want to use the `teletk` R package or build package documentation from sources.

3. After the download is finished, click on the downloaded file, then follow the instructions on the installation pop-up window.

Download, install RStudio

To download RStudio, go to rstudio.com/products/rstudio/download/#download and download the version for macOS.

2. Download RStudio Desktop. Recommended for your system:

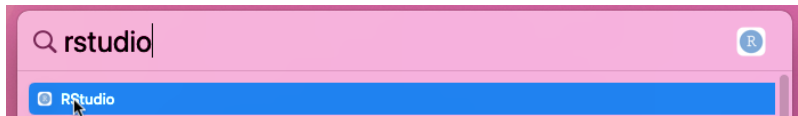


DOWNLOAD RSTUDIO FOR MAC

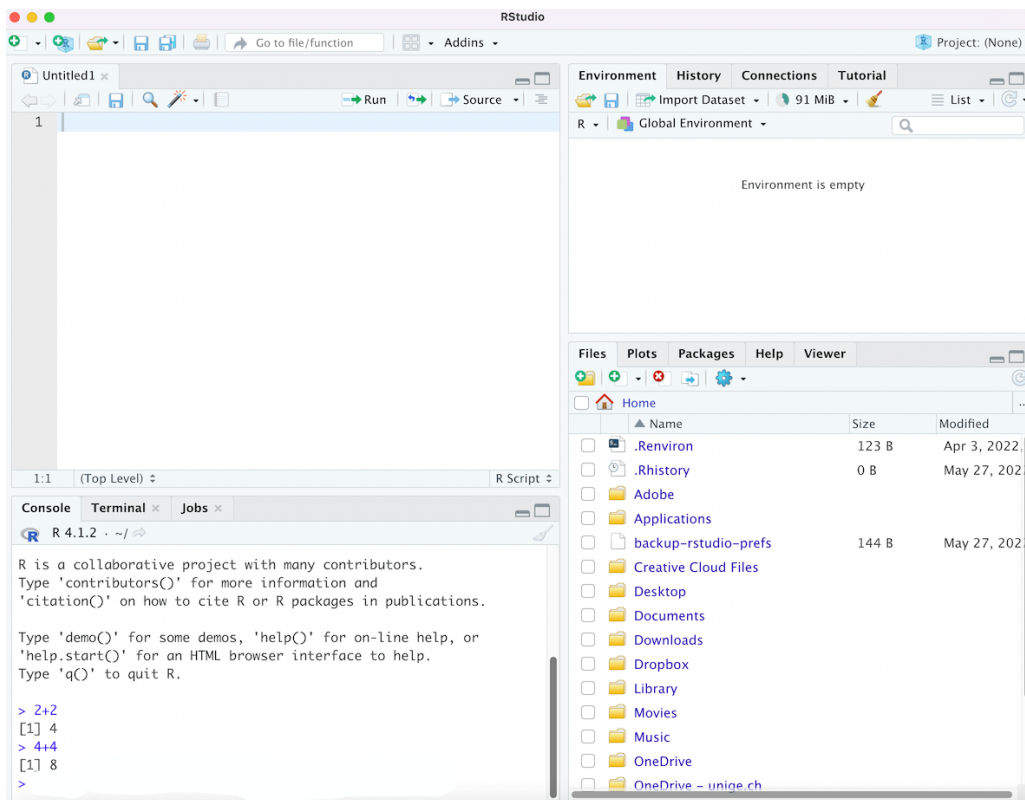
2022.02.0+443 | 217.18MB

After the download is finished, click on the downloaded file and follow the installation instructions.

Once installed, RStudio can be opened like any application on your computer: Press Command + Space to open Spotlight, then search for “rstudio”. Click to open the app.

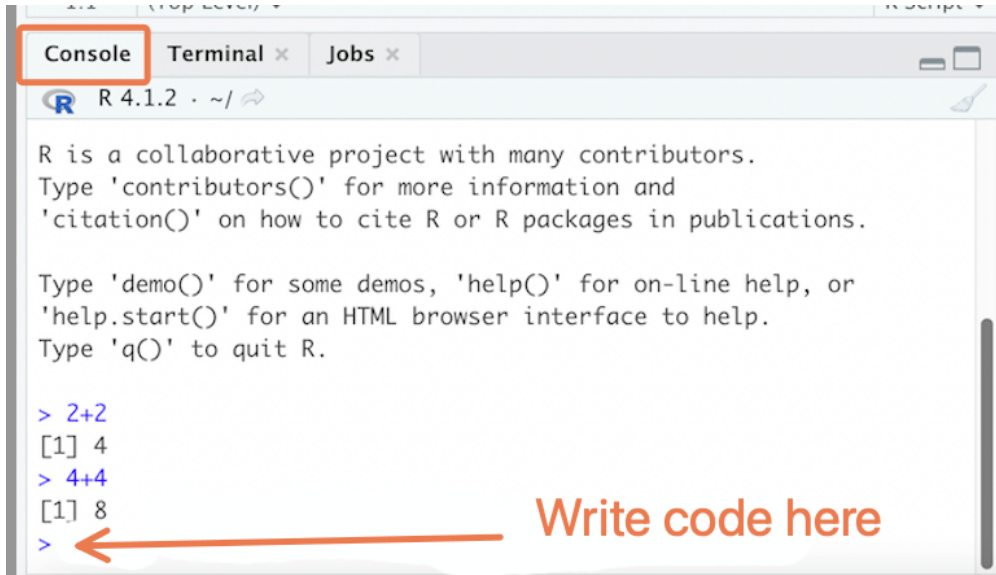


You should see a window like this:



This is RStudio, your new home for a long time to come!

You can start using R by typing code into the “console” pane on the left:



```
R 4.1.2 · ~/  
```

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

```
> 2+2  
[1] 4  
> 4+4  
[1] 8  
>
```

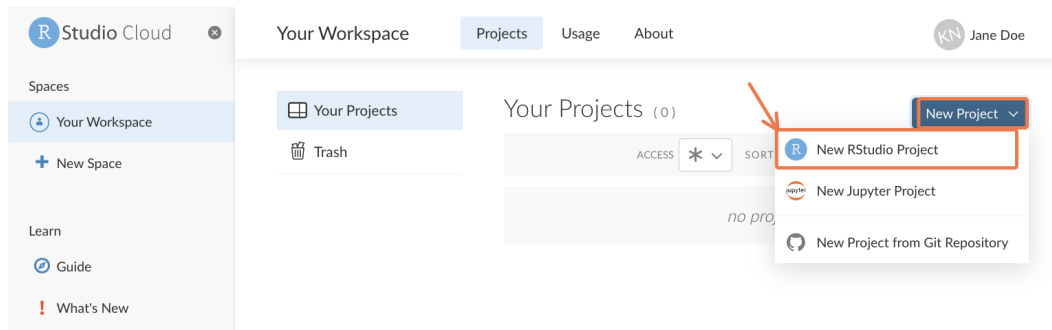
Write code here

Try using R as a calculator here; type `2 + 2` and press Enter.

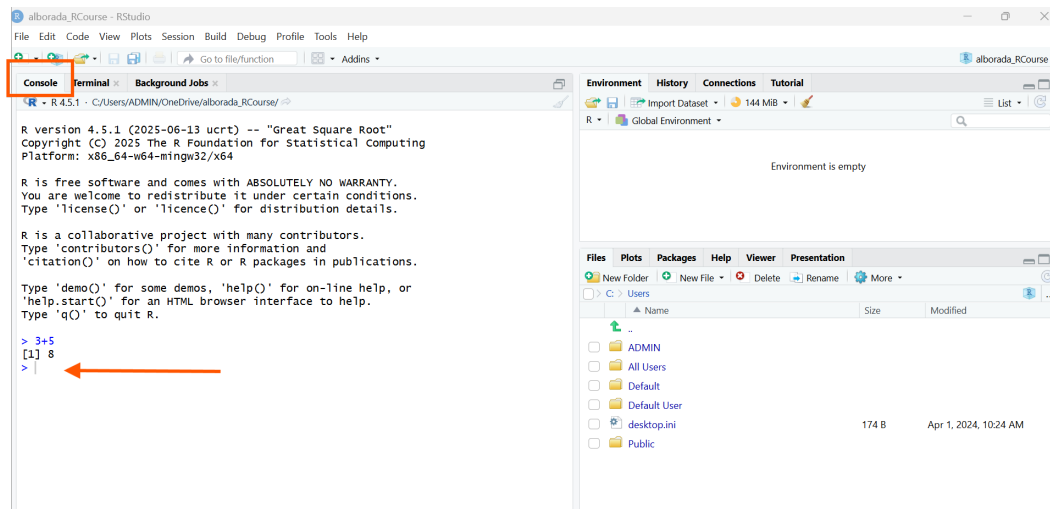
RStudio on the cloud

If you'll be working on the cloud, follow the steps below:

1. Go to the website rstudio.cloud and follow the instructions to sign up for a free account. (We recommend signing up with Google if you have a Google account, so you don't need to remember any new passwords).
2. Once you're done, click on the “New Project” icon at the top right, and select “New RStudio Project”.

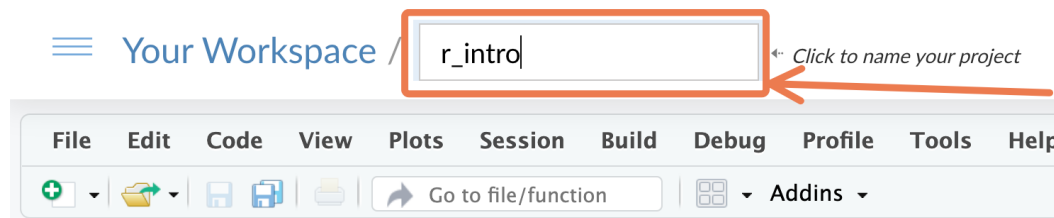


You should see a screen like this:



This is RStudio, your new home for a long time to come!

At the top of the screen, rename the project from “Untitled Project” to something like “r_intro”.



You can start using R by typing code into the “console” pane on the left:


```
R version 4.2.0 (2022-04-22) -- "Vigorous Calisthenics"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Session restored from your saved work on 2022-May-29 08:52:07 UTC (41
minutes ago)
> 2 + 2
```

Try using R as a calculator here; type `2 + 2` and press Enter.

That's it; you're ready to roll. Whenever you want to reopen RStudio, navigate to rstudio.cloud,

Proceed to the “wrapping up” section of the lesson.

- RStudio cloud will only give you 25 free project hours per month. After that, you will need to upgrade to a paid plan.
- If you think you'll need more than 25 hours per month, you may want to avoid this option.

